Richard White Ph.D.

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Education

2002

PhD

University of London

1997

BA, Natural Sciences

Clare College,

University of Cambridge

Publications

0000-0003-1842-412X

Senior Bioinformatician

Experienced bioinformatician with strong science background. I am adept at producing visualisations that convey complex information clearly. I am curious about data and how information can be presented for maximum understanding. Strong proponent of open science and open data.

Technical Skills

10 years' experience writing python for data analysis, which has contributed to high quality peer-reviewed scientific publications and insight into biological processes and clinical syndromes

10+ years' experience using R for data wrangling, statistical analysis and visualisation to create figures for scientific publications to convey simple stories from complex data

Using Git and GitHub for version control and code collaboration and review (link in sidebar)

Writing R Shiny apps to make interactive visualisations to help make research outputs available to other scientists

Experience using high-performance computing systems (LSF, SLURM, UGE) and Singularity containers, which is applicable to cloud computing systems

Experience using the Unix command line, Perl (10+ years), databases (MySQL, SQLite) and Nextflow workflow language

Employment History

Senior Bioinformatician

Queen Mary University of London (2022-present), Busch Lab

As part of a project to create gene expression correlation networks, I learnt to use Nextflow to create an analysis pipeline, which allowed me to rapidly prototype solutions and iterate, to speed up the development process. I produced a simple web app using the Python Flask framework, to automate and speed up design of experimental reagents, building on someone else's Python module. I am currently mentoring PhD students, teaching them lab techniques and reviewing their code. I have enjoyed building effective working relationships with the students and with their supervisor.

University of Cambridge (2018-2022), Busch Lab

Worked on the 'Deciphering the Mechanisms of Developmental Disorders' project - an international consortium of scientists with many different specialties. I was the joint-lead on our section of the analysis programme, resulting in a publication in the journal *Nature Communications*.

I was a course organiser on the EMBL-EBI course, '**Bioinformatics and functional genomics in zebrafish (2019, 2022)**'. I planned, designed and delivered modules on Introductions to R, Data Visualisation with R and Cytoscape for Visualisation.

Expertise

- Data Processing
- Data Visualisation
- Machine Learning
- Statistical Analysis
- Relational Databases
- Course Design
- Course Materials
- Reserach Strategy
- Experimental Design
- Reports
- Presentation
- Publications

Wellcome Trust Sanger Institute, Cambridge (2016-2018)

Worked on multiple collaborative projects. One was an analysis of gene-expression data from Shigella infections in zebrafish. It involved researchers in two different labs with knowledge in infectious diseases, but not informatics. This required me to present the analysis to people with a non-informatics background, but also to challenge them to add their domain expertise to produce the correct interpretation.

Another collaboration was a clinical project on a rare disease syndrome, using a zebrafish model of the disease. It involved successive rounds of analysis, feedback and re-analysis and resulted in a new understanding of possible reasons for the different outcomes to treatment that patients experience.

Computer Biologist

Zebrafish Mutation Project, Wellcome Sanger Institute, Cambridge (2011-2016)

Part of an academic team to produce a high-quality timecourse of zebrafish gene expression data. We worked with another team (Expression Atlas) to make the data publicly available in a user-friendly interface. This enabled researchers to look up the expression of specific genes that they are interested in and drastically increased the reach and impact of the project. I was lead author on the resulting highly-cited publication.

I used machine-learning approaches (support vector machines and neural networks) on a project to improve the prioritisation of predicted genetic variants, thereby increasing the output of the analysis pipeline.

As an author on a highly cited paper in the prestigious journal Nature, I worked as part of a team to quickly produce a credible reasoned response to the reviewers' comments. This required decision-making among 20 co-authors in a limited timeframe.

Postdoctoral Fellow

Wellcome Sanger Institute, Cambridge (2007-2011)

Worked on a project to infer gene regulation from gene expression data. This project sparked my passion for big data projects and was the driver for my switch to bioinformatics.

Postdoctoral Scholar

University of California, Irvine, USA (2002-2007)

Led a project on the role of retinoic acid during development, in which we collaborated with academics in the Maths department to use modelling to produce testable predictions for our experimental work. As part of this, I was selected to present a talk at the Society for Development Biology meeting, a large international academic conference.

Personal

I have been head coach (ECB Core Coach/Level2) of the Pirton cricket club junior section for 3 years. This involves organising league fixtures, planning and running coaching sessions and running league games. I am committed to encouraging participation for all.